David Nardi

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SUMMARY

AI Engineer with an Engineering Management background, specialized in designing and deploying scalable AI training systems and MLOps workflows. Successfully reduced model training downtime by 60% and enabled reproducibility for LLM studies through automated pipelines. Looking to drive impact at the intersection of research and scalable production.

EDUCATION

MSc in Artificial Intelligence Engineering. University of Florence, Italy — 2023 - Apr 2026

- **Thesis**: Specialized optimization algorithms for learning under heavy-tailed class imbalance with applications to LLMs pre-training.
- **GPA:** 29/30
- **Relevant Coursework:** Deep Learning, NLP & LLMs, Generative & Multimodal AI, Computer Vision, Reinforcement Learning, Machine Learning Security, Optimization, Statistics.
- **Activities:** Student representative; Member of AI Engineering Degree Review Committee.

BSc in Management Engineering (IT Track). University of Florence, Italy — 2019 - 2023

- **Thesis**: Machine learning applications to industrial quality control.

TECHNICAL SKILLS

Languages: Python, SQL, R, Bash
AI/ML Frameworks: PyTorch, HuggingFace Ecosystem, FAISS, Scikit-learn, spaCy, Optuna
MLOps & Infrastructure: Slurm, Git, Linux, WandB, CometML
Speaking: Italian (Native), English (C1)

EXPERIENCE

AI Research Intern. Mathematics Institute, LMU Munich, Germany — Sep 2025 - Feb 2026

Research focus: understanding why the Adam optimizer effectively pre-trains language models.

- Build a modular, scalable **Python/PyTorch** training pipeline integrating **Optuna** and **CometML**, achieving 100% experimental reproducibility across 3 core LLM pre-training studies.
- Designed distributed LLM pre-training workflows on an **8-GPU Slurm cluster**, reducing training downtime by eliminating manual interventions through automation.
- Identified actionable research directions on **deep learning optimization** with insights from 10+ papers by conducting a structured literature review on the SGD-Adam performance gap in LLM pre-training.
- Presented research findings to an international audience (20+ attendees), incorporating expert feedback into experimental design.

Co-Head of Business Operations. JEFLO, Junior Enterprise Florence, Italy — Mar 2024 - Jun 2025

Student-run consulting network providing hands-on entrepreneurial experience through real-world projects.

- Reduced **project planning** effort by 40% by developing an internal management system. Broadened services scope with 3 new projects by training the team (10 consultants) on **dashboarding** for a total of 5 hours.
- Selected as one of 500+ participants Europe-wide for a **4-days conference** to represent the organization.

TECHNICAL PROJECTS

Slurm jobs monitor

- Engineered and deployed a real-time Slurm job-monitoring service using **Python**, **Slurm**, **HuggingFace Transformers** (log-summarization pipeline), enabling researchers to receive periodic updates and instant notifications on state transitions (`pending` → `running` → `failed/completed`).
- Reduced manual **cluster checks** by 60% and cut average detection time for `failed` jobs from 10 minutes to milliseconds, improving iteration speed and resource utilization across 10 active users and 50 daily jobs.

Argument Mining with NLP

- Led **fine-tuning and model selection** across 6 DistilBERT and SBERT variants for argument-component detection, achieving a 20% F1 improvement over baseline.
- Built and deployed an end-to-end **inference pipeline** processing 50+ medical research articles, enabling retrieval of high-argumentative-content exploiting custom scoring metrics.